CARAT: Class-Aware Robust Aggregation via Hidden Task Certificates for Poisoning-Resilient Federated Learning

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Abstract

Federated learning servers aggregate client updates without seeing client data, which leaves training exposed to both untargeted model poisoning and targeted backdoor attacks. Most robust aggregators rely on update geometry, coordinate-wise robust statistics, or alignment with a single trusted reference direction. These signals weaken when benign clients are heterogeneous and adaptive attackers stay close to the benign update manifold. We propose **CARAT**, a class-aware robust aggregation rule built on *hidden task certificates*. Given a small labeled server-side reference set, CARAT samples multiple hidden, class-balanced micro-tasks and scores each client update by the loss reduction it achieves across them. CARAT combines these task-conditioned scores with robust clipping, common-radius evaluation, coordinate-rank anomaly penalties, and capped-simplex weight optimization with temporal regularization. The result is an aggregator that favors updates that remain useful across diverse semantic slices while limiting the influence of structurally extreme clients. We also define an evaluation protocol for CARAT that emphasizes class-rich benchmarks, separate untargeted and backdoor threat models, matched trusted-data budgets, and class-sensitive metrics such as worst-class accuracy and attack success rate.

1 Introduction

Federated learning (FL) lets many clients train a shared model without centralizing raw data [1–3]. This decentralization is attractive in privacy- and governance-sensitive settings, but it leaves the server in an asymmetric position: the global model must be updated from client messages that the server cannot directly verify. The aggregation rule therefore becomes a security-critical part of the FL system. Malicious clients can slow convergence, distort the decision boundary, or implant triggers that activate attacker-chosen behavior at test time [4–11].

The central difficulty is that benign heterogeneity and malicious behavior can look similar from the server's perspective. Under non-i.i.d. partitions, benign clients can differ substantially in update direction, norm, sparsity, and class emphasis [12–15]. An unusual update may therefore reflect either a harmful client or a legitimately different local distribution. This ambiguity weakens defenses based on coarse notions of centrality. Pairwise-distance methods can mistake benign dispersion for adversarial deviation, coordinate-wise robust statistics discard cross-coordinate structure, and trusted-anchor approaches compress all auxiliary supervision into a single global reference direction [16–23]. Such signals are often too coarse to expose attacks whose effect is concentrated on a subset of classes or semantic regions.

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CARAT addresses this problem by shifting validation from geometric similarity to task-conditioned utility. The server maintains a small labeled reference set and repeatedly instantiates hidden, class-balanced micro-tasks from it. Each client update is scored by the loss reduction it induces across these tasks, producing a certificate matrix that measures whether the update is broadly useful across semantic slices of the problem instead of merely close to the geometric center of the client population. CARAT then combines this signal with robust clipping, common-radius evaluation, coordinate-rank anomaly penalties, and capped-simplex weight optimization with temporal regularization.

This design targets FL deployments in which a modest server-side audit set is realistic, such as cross-silo or institutionally managed settings. It is not a fully data-free defense. Within that setting, CARAT aims to preserve benign diversity while reducing the influence of updates that are semantically harmful or structurally extreme. Our main contributions are as follows.

- We propose CARAT, a class-aware robust aggregation rule that validates client updates through hidden-task certificates constructed from server-side labeled reference data, replacing one-shot geometric trust signals with multi-task utility checks.
- We formulate aggregation as a capped-simplex optimization problem that combines worst-task utility across hidden tasks, coordinate-rank anomaly suppression, and temporal regularization, yielding a robust weighting mechanism compatible with heterogeneous FL.
- We define an evaluation protocol for CARAT centered on class-rich benchmarks, separate untargeted and backdoor suites, matched trusted-data budgets, and class-sensitive metrics, so that the empirical study directly tests the semantic robustness claims of the method.

2 Related Work

Poisoning attacks. Modern FL attacks are designed to remain effective under server-side defenses. Early work already showed that even constrained adversaries can poison the global model while remaining relatively stealthy [4–6]. For untargeted degradation, attacks such as ALIE, MinMax, and MinSum craft malicious updates that evade simple geometric or statistical filters while still harming convergence or accuracy [6, 7]. For targeted manipulation, model-replacement backdoors, distributed trigger construction, tail attacks, and durable attacks such as Neurotoxin show that an update can remain globally plausible while implanting highly specific malicious behavior [8–11]. Surveys make the broader point explicit: once the server cannot inspect raw client data, poisoning and backdoor attacks become a systems concern rather than an edge case [3, 24].

Robust aggregation under heterogeneity. Classical Byzantine-robust aggregation rules infer benignness from centrality. Krum and related geometry-based methods favor updates that remain close to the client population under pairwise distances [16, 17], while coordinate-wise median and trimmed mean suppress marginal extremes independently [18]. Later variants extend this family through suspicion scoring, regularized aggregation, coding-based redundancy, and geometric-median aggregation [19–22, 25]. These methods are attractive because they require little server-side prior knowledge. Under heterogeneous data, however, they can still confuse legitimate benign dispersion with malicious deviation. This limitation has motivated methods that address heterogeneity more directly through optimization stabilization, resampling, bucketing, local-SGD analysis, gradient splitting, nearest-neighbor mixing, label-skew-aware aggregation, and entropy-based public-data guidance [13–15, 26–32].

Trusted-data and backdoor defenses. A third line of work augments aggregation with trusted data, history, or representation-level inspection. In collaborative learning, AUROR anticipated update screening for poisoning resistance [33]. In FL, FLTrust introduces a trusted server anchor, FoolsGold and FLDetector rely on historical similarity diagnostics, and FLARE, FLAME, DeepSight, CRFL, and FLIP use latent-space inspection, clustering, clipping, smoothing, or reverse-engineering-based backdoor mitigation [23, 34–40]. Recent work also highlights the cost of robustness itself, including clipping design and the tradeoff between Byzantine resilience and clean accuracy [41, 42]. CARAT is closest to this broader defense-and-detection family, but it uses a different validation principle: instead of reducing trust to a single anchor direction, geometric similarity, or post-hoc clustering signal, it asks whether a client update remains useful across multiple hidden semantic sub-tasks.

3 Method

3.1 Formulation and Assumptions

We consider synchronous FL with n clients and a global parameter vector $w_r \in \mathbb{R}^d$ at round r. At each round, client i returns a message u_i^r, which the server maps to a vector-form update $\Delta_i^r \in \mathbb{R}^d$. For FedAvg, Δ_i^r is the client model delta relative to the current global model; for gradient-style optimizers, it is the client update itself. The server must aggregate $\{\Delta_i^r\}_{i=1}^n$ into a single global step while a fraction ρ of clients may be malicious and coordinated.

CARAT assumes the server has access to labeled reference data $\mathcal{D}^{\text{ref}}$. From this data it constructs a probe pool $\mathcal{P}$ and repeatedly samples hidden, class-balanced validation tasks. This assumption is stronger than fully data-free robust aggregation and should be treated as part of the threat model, not as an incidental implementation detail. CARAT uses the auxiliary data not to define a single trusted direction, but to generate multiple hidden semantic checks against which candidate client updates can be evaluated.

The objective is to assign aggregation weights that preserve benign diversity while suppressing updates that are semantically harmful or structurally anomalous. This distinction matters in non-i.i.d. FL, where benign clients may differ substantially in direction, norm, and class emphasis. Geometric dispersion alone is therefore insufficient. CARAT divides each round into three stages: common-geometry evaluation, task-aware and structural scoring, and robust weight optimization. Evaluation is normalized across clients, while the final server step remains conservative because it is computed from clipped updates.

3.2 Method Details

At round r, CARAT first converts client messages into vector-form deltas and computes their norms. Before any semantic evaluation, the server robustly clips client updates so that a small number of large-magnitude messages cannot dominate the comparison stage. Let $\|\Delta_i\|_2$ denote the norm of client i's update. In our implementation, the clipping threshold τ_r is estimated by the MAD-based rule

$$\tau_r = \text{median}(\{\|\Delta_i\|_2\}) + k \cdot 1.4826 \cdot \text{MAD}(\{\|\Delta_i\|_2\}), \tag{1}$$

and the clipped update is

$$\bar{\Delta}_i = \min\left(1, \frac{\tau_r}{\|\Delta_i\|_2 + \varepsilon}\right) \Delta_i. \tag{2}$$

CARAT then rescales each clipped update to a common evaluation radius q_r:

$$\tilde{\Delta}_i = q_r \frac{\bar{\Delta}_i}{\|\bar{\Delta}_i\|_2 + \varepsilon}, \tag{3}$$

where q_r is chosen as a quantile of the positive clipped norms. This normalization removes pure scale advantages during hidden-task comparison. The normalized vectors are used only for evaluation; the final model update is still computed from the clipped vectors. CARAT uses normalization to create a fair scoring geometry, not to define the actual optimization step taken by the server.

After preprocessing, CARAT samples T hidden tasks from the probe pool $\mathcal{P}$, with each task constructed to be approximately class-balanced. This is what makes the method class-aware. A malicious update can remain globally plausible while degrading only a subset of classes or semantic regions. Repeated balanced hidden tasks make these localized failures harder to hide. For each hidden task t, let $\ell_t(w)$ denote the average cross-entropy loss of model w on that task. CARAT defines a *hidden-task certificate*

$$c_{i,t} = \ell_t(w_r) - \ell_t(w_r + \tilde{\Delta}_i). \tag{4}$$

Here, "certificate" denotes an internal task-level utility score, not a formal certified-robustness guarantee. Positive values indicate that client i improves task t relative to the current global model.

Stacking these scores yields a certificate matrix $C \in \mathbb{R}^{n \times T}$. Because different tasks may have different intrinsic difficulty and baseline loss scale, CARAT robustly standardizes each column of C before optimization. This preserves within-task ordering while making scores comparable across hidden slices. The resulting matrix gives the server a richer signal than client-to-client similarity

alone: a client should be trusted because it remains broadly useful across hidden tasks, not merely because it lies near the geometric center of the client population.

Task utility alone is insufficient, because a malicious client may overfit the probe tasks or align with part of the semantic signal while remaining structurally extreme in parameter space. CARAT therefore augments hidden-task certificates with a coordinate-rank anomaly score. For a sampled coordinate set $S \subseteq \{1, \ldots, d\}$, it ranks clients coordinate-wise and measures the average normalized distance from the center rank:

$$r_i^{(S)} = \frac{1}{|S|} \sum_{j \in S} \left| \frac{\mathrm{rank}_{i,j}}{n-1} - \frac{1}{2} \right|. \tag{5}$$

CARAT repeats this computation over several random coordinate subsets and aggregates the resulting values robustly. A client receives a large anomaly score if its update remains persistently extreme across many coordinates rather than on a few isolated entries. After robust standardization, only positive deviations are retained as penalties. The signal is intentionally asymmetric: suspicious extremeness is penalized, whereas unusual centrality is not explicitly rewarded. Together, the certificate matrix and rank penalty separate two notions that are often conflated in robust aggregation: whether an update is useful and whether it is structurally atypical.

3.3 Weight Optimization and Aggregation

CARAT maps these signals to aggregation weights by solving a constrained optimization problem on the capped simplex

$$\mathcal{A}(\hat{\rho}) = \left\{ \alpha \in \mathbb{R}^n : \alpha_i \geq 0,\ \sum_{i=1}^{n} \alpha_i = 1,\ \alpha_i \leq \frac{1}{(1-\hat{\rho})n} \right\}, \tag{6}$$

where $\hat{\rho}$ is an estimate of the attacker fraction. The cap prevents any single client from dominating the aggregate and encodes a conservative trust budget at the weight level: even when a client appears useful, it cannot absorb arbitrary influence within one round.

Let $r \in \mathbb{R}^n$ denote the positive rank penalties and let α^{prev} be the previous round's weights. CARAT solves

$$\max_{\alpha \in \mathcal{A}(\hat{\rho})} \ \operatorname*{softmin}_{\beta}(C^\top \alpha) - \lambda_r r^\top \alpha - \lambda_p \|\alpha - \alpha^{\mathrm{prev}}\|_2^2, \tag{7}$$

where

$$\operatorname*{softmin}_{\beta}(s_1, \ldots, s_T) = -\frac{1}{\beta} \log \left(\frac{1}{T} \sum_{t=1}^{T} e^{-\beta s_t} \right). \tag{8}$$

The first term is a smooth approximation to worst-task utility: large β places more emphasis on the weakest hidden tasks, while finite β preserves optimization stability. The second term penalizes structurally extreme clients, and the third term regularizes the solution toward the previous round to reduce unnecessary oscillation. Eq. (7) therefore implements a simple decision rule: a client should retain influence only if it remains useful across hidden tasks and does not repeatedly appear as an outlier in rank space.

After optimization, CARAT aggregates the clipped updates:

$$\Delta_{\mathrm{agg}} = \sum_{i=1}^{n} \alpha_i \bar{\Delta}_i. \tag{9}$$

This is the final server step. The aggregate inherits the robustness benefits of clipping while using weights learned from the richer evaluation geometry defined above. In practice, the main additional computation comes from evaluating n candidate updates over T hidden tasks and computing rank penalties on coordinate subsamples. Probe-model forward passes dominate, while the ranking and optimization stages remain tunable through explicit budget parameters.

A compact round-level algorithm summary is provided in Appendix A. Formal properties of the normalization step, capped-simplex feasibility, softmin behavior, temporal regularization, and per-round complexity are deferred to Appendix B.

4 Experimental Protocol

4.1 Evaluation Objectives

CARAT is intended for settings where the server has a small labeled audit set and where failures may concentrate on particular classes rather than appearing only as global geometric outliers. The empirical study therefore has four objectives: measure CARAT's clean-utility cost in the absence of attack, test robustness to untargeted model poisoning under increasing client heterogeneity, evaluate robustness to class-selective and backdoor attacks, and quantify the tradeoff between robustness, server-reference budget, and runtime overhead. The full benchmark matrix, appendix-only stress tests, and ablation scope are reported in Appendix C.

4.2 Benchmark Design

Datasets and models. The primary benchmark is CIFAR-100 with ResNet18 [43]. CIFAR-100 offers a class-rich label space in which hidden semantic slices are meaningful and class-selective failures are easier to expose than on low-cardinality datasets. We complement it with TinyImageNet, again using ResNet18, to test whether the method scales to a larger class vocabulary while keeping the backbone fixed. To anchor the method in a more application-oriented setting, we also include CHMNIST with the same backbone in the appendix, where class-selective degradation has a clearer domain interpretation. CIFAR-10 is retained only as an appendix sanity check rather than a primary source of evidence, because a ten-class benchmark is too coarse to serve as the main validation vehicle for a class-aware defense. Keeping the model family fixed across visual datasets helps attribute observed differences to aggregation rather than to backbone changes.

Untargeted poisoning suite. For untargeted model poisoning, the main threat models are ALIE [7], MinMax [6], and MinSum [6]. These attacks represent stealthy optimization-based adversaries that try to stay near the benign update manifold while degrading global learning. The primary untargeted baselines are Mean, TrimmedMean [18], MultiKrum [16], and FLTrust [23]. FangAttack [5] is treated as an appendix stress test rather than a core main-text attack because it is especially tied to Krum-style geometry.

Targeted and backdoor suite. The second core suite targets semantic corruption. We focus on ModelReplacement-style backdoors [8], DBA [9], and Neurotoxin [11], because these attacks directly test whether CARAT's hidden tasks can expose updates that preserve apparent global plausibility while damaging specific semantic regions. The main backdoor baselines are Mean, FLTrust, FLAME [37], DeepSight [38], and CARAT. BadNets-style triggers can be retained in the appendix as a sanity check, but they are not the main stress case because they are typically easier to detect than adaptive distributed or persistent attacks.

Shared protocol. Across the primary suites, we use 20 clients with full participation, 300 global rounds, 4 local epochs, and learning rate 0.05. To probe heterogeneity, we evaluate i.i.d. partitions together with Dirichlet non-i.i.d. regimes $\alpha \in \{0.5, 0.1\}$. The main adversarial setting uses an attacker ratio of 20%, while 40% is reserved for appendix stress tests. Clean-reference experiments are reported separately from attacked experiments instead of being inserted as a row inside attack tables. Unless otherwise stated, reported means and standard deviations are computed over 3–5 random seeds.

4.3 Metrics and Fairness Controls

Because CARAT is explicitly class-aware, the evaluation must report more than aggregate clean accuracy. For clean utility, we report test accuracy and test loss under no attack. For robustness, we additionally report macro-average accuracy and worst-class accuracy on class-rich benchmarks so that localized failures remain visible. For backdoor experiments, attack success rate (ASR) is always paired with clean accuracy rather than reported in isolation. Since Proposition B.7 identifies probe-model evaluation as the main extra cost, we also report runtime per round and total defense overhead.

Fairness requires explicit control of server-side supervision. CARAT uses labeled server reference data to instantiate hidden tasks, while FLTrust uses trusted data to define an anchor update; these

Table 1: Experimental protocol for evaluating CARAT.

Component	Design choice
Primary benchmark	CIFAR-100 with ResNet18
Supplementary benchmarks	TinyImageNet with ResNet18; CHMNIST with ResNet18 (appendix); CIFAR-10 only as appendix sanity check
Clients and participation	20 total clients with full participation each round
Optimization schedule	300 communication rounds, 4 local epochs, learning rate 0.05
Heterogeneity settings	i.i.d.; Dirichlet non-i.i.d. with $\alpha \in \{0.5, 0.1\}$
Main adversarial fraction	20% malicious clients
Stress adversarial fraction	40% malicious clients (appendix)
Main untargeted attacks	ALIE, MinMax, MinSum
Main backdoor attacks	ModelReplacement, DBA, Neurotoxin
Untargeted baselines	Mean, TrimmedMean, MultiKrum, FLTrust, CARAT
Backdoor baselines	Mean, FLTrust, FLAME, DeepSight, CARAT
Primary reported metrics	Clean accuracy, clean loss, macro accuracy, worst-class accuracy, ASR, runtime per round
Fairness controls	Matched labeled server-reference budget between CARAT and FLTrust; clean-reference runs reported separately from attacked runs
Repetitions	3–5 seeds per configuration

methods should therefore be compared under a matched labeled-data budget. In the main text, CARAT and FLTrust are evaluated with the same reference budget. The appendix then sweeps this budget to distinguish gains due to the evaluation principle from gains due only to more public data. Likewise, we do not use a single dataset-independent probe-pool size across all benchmarks: the reference budget is parameterized per class so that class-rich datasets are not artificially underprovisioned.

4.4 Reporting Structure

The evaluation is organized around three result groups: a clean-utility comparison, an untargeted robustness comparison under increasing heterogeneity, and a backdoor comparison emphasizing class-sensitive metrics and ASR. Appendix C records the full benchmark suites, stress tests at higher attacker ratios, scaling studies, trusted-data budget sensitivity, runtime breakdowns, and CARAT-specific ablations.

5 Conclusion

We presented CARAT, a class-aware robust aggregation rule that validates client updates through hidden task certificates rather than relying only on geometric centrality or a single trusted reference direction. By scoring updates across multiple hidden, class-balanced semantic slices and combining those scores with rank-space anomaly suppression and capped-simplex weight optimization, CARAT is designed to preserve benign heterogeneity while reducing the influence of semantically harmful or structurally extreme clients.

The method is most appropriate in FL deployments where the server can maintain a modest labeled audit set and where class-selective failures matter. In that regime, the relevant empirical evaluation centers on class-rich benchmarks, separate untargeted and backdoor threat models, class-sensitive metrics such as macro and worst-class accuracy, and explicit controls for trusted-data budget and runtime overhead. CARAT's central claim is not only that it rejects large outliers, but that it favors updates that remain consistently useful across hidden semantic slices even when attacks are adaptive and benign clients are heterogeneous.

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Algorithm 1 CARAT server aggregation at round r

Require: Current global model w_r, client messages $\{u_i^r\}_{i=1}^n$, previous-round weights α^{prev}, probe pool $\mathcal{P}$, CARAT hyperparameters
Ensure: Aggregated update Δ_{agg}, updated temporal prior α^{prev}

```
if P is uninitialized then
    Build a balanced probe pool from the server-visible training dataset
end if
if r = 1 then
    α^prev ← (1/n) 1
end if
for i = 1 to n do
    Convert u_i^r into a vector-form delta Δ_i^r
end for
Compute the clipping threshold τ_r from {‖Δ_i^r‖_2}_{i=1}^n
for i = 1 to n do
    Δ̄_i^r ← min(1, τ_r / (‖Δ_i^r‖_2 + ε)) Δ_i^r
end for
Choose the evaluation radius q_r from a quantile of positive clipped norms
for i = 1 to n do
    Δ̃_i^r ← q_r Δ̄_i^r / (‖Δ̄_i^r‖_2 + ε)
end for
Sample T class-balanced hidden tasks from P with replacement/subset fallback if strict balanced
sampling fails
for t = 1 to T do
    Evaluate the baseline loss ℓ_t(w_r)
end for
for i = 1 to n do
    for t = 1 to T do
        c_{i,t} ← ℓ_t(w_r) − ℓ_t(w_r + Δ̃_i^r)
    end for
end for
Robustly standardize certificate columns to obtain C^std
Compute rank penalties r from repeated coordinate subsamples of {Δ̄_i^r}_{i=1}^n
α^(0) ← α^prev
for k = 0 to K − 1 do
    z ← (C^std)^⊤ α^(k)
    π ← softmin task weights induced by z and β
    g ← C^std π − λ_r r − 2λ_p (α^(k) − α^prev)
    α^(k+1) ← Π_{A(ρ̂)}(α^(k) + ηg)
end for
α* ← α^(K) and α^prev ← α*
Δ_agg ← Σ_{i=1}^n α_i* Δ̄_i^r
return Δ_agg, α^prev
```

A Algorithmic Details

A.1 Round-Level Pseudocode

Algorithm 1 summarizes one CARAT server round in standard pseudocode form.

A.2 Implementation Notes

Inputs. Current global model w_r, client messages $\{u_i^r\}_{i=1}^n$, previous-round weights α^{prev}, server-visible training dataset, and CARAT hyperparameters.

Persistent state. CARAT maintains two pieces of state across rounds: a balanced probe pool sampled at initialization from the server-visible training dataset, and the previous-round weight vector α^{prev}, which serves as the temporal prior for the current round.

Initialization. At the first round, the implementation initializes α^{prev} to the uniform vector. This keeps the initial round unbiased across clients while still allowing later rounds to exploit temporal consistency once CARAT has observed several optimization trajectories.

Reference-pool construction and hidden-task sampling. Let $\mathcal{P}_c \subseteq \mathcal{P}$ denote the subset of probe examples with class label c. A hidden task is constructed by sampling a small number of examples from each available class pool and concatenating them into one balanced micro-task. If the desired number of samples per class is not available without replacement, CARAT falls back to replacement and, if needed, to a random subset of the probe pool rather than aborting the round. This fail-soft design preserves the intended semantic diversity of the task sampler while ensuring that the server always obtains a well-defined certificate matrix.

Robust score calibration. For notational economy, the main text reuses C after standardization. In implementation terms, the raw certificate matrix is first transformed column-wise into

$$C^{\mathrm{std}}_{i,t} = \frac{c_{i,t} - \mathrm{median}_j(c_{j,t})}{1.4826 \cdot \mathrm{MAD}_j(c_{j,t}) + \varepsilon}.$$

This removes task-specific scale effects while preserving within-task ordering by Proposition B.4. Rank penalties are calibrated similarly. For each coordinate subsample S_s, let

$$\rho_i^{(s)} = \frac{1}{|S_s|} \sum_{j \in S_s} \left| \frac{\mathrm{rank}_{i,j}}{n-1} - \frac{1}{2} \right|.$$

CARAT then computes a standardized rank score

$$z_i^{(s)} = \frac{\rho_i^{(s)} - \mathrm{median}_\ell(\rho_\ell^{(s)})}{1.4826 \cdot \mathrm{MAD}_\ell(\rho_\ell^{(s)}) + \varepsilon},$$

and aggregates subsamples through

$$r_i = \max\left(0, \mathrm{median}_s\, z_i^{(s)}\right).$$

As a result, a client is penalized only when it is consistently extreme across repeated coordinate views, not because of a single noisy subsample.

Projected optimization details. Again writing C for the standardized certificate matrix, CARAT initializes the optimizer with $\alpha^{(0)} = \alpha^{\mathrm{prev}}$ after the first round, and with the uniform vector at initialization. If

$$z(\alpha) = C^\top \alpha, \qquad \pi_t(\alpha) = \frac{\exp(-\beta z_t(\alpha))}{\sum_{\ell=1}^{T} \exp(-\beta z_\ell(\alpha))},$$

then the ascent direction used in projected updates is

$$g(\alpha) = C\pi(\alpha) - \lambda_r r - 2\lambda_p(\alpha - \alpha^{\mathrm{prev}}).$$

With optimizer step size η, one iteration takes the form

$$\alpha^{(k+1)} = \Pi_{\mathcal{A}(\hat{\rho})}\left(\alpha^{(k)} + \eta g(\alpha^{(k)})\right).$$

The projection maintains feasibility throughout optimization, so every intermediate iterate already satisfies the nonnegativity, unit-sum, and per-client cap constraints described in the main text.

Round invariants and failure handling. Two implementation choices are important for robustness. First, the optimizer never leaves the capped simplex, so every intermediate and final weight vector remains feasible. Second, probe-task construction is fail-soft rather than fail-stop: if strict class-balanced sampling is impossible, the code falls back to replacement or to a random subset, ensuring that the round can still proceed with a well-defined certificate matrix.

Outputs. Aggregated server update, updated temporal prior α^{prev}, and diagnostic logs including learned weights, task scores, softmin focus, rank penalties, evaluation radius, clipping threshold, and the identities of the hidden probe tasks used in the current round.

Practical diagnostics. The most useful round-level diagnostics for debugging and empirical interpretation are the learned weights, the weakest-task values emphasized by the softmin, the clip threshold τ_r, the evaluation radius q_r, and the final rank penalties r_i. In successful runs, these quantities help distinguish three qualitatively different cases: benign but diverse clients that retain moderate weight, clearly harmful clients that lose certificate support across multiple tasks, and structurally extreme clients whose influence is reduced primarily by the rank penalty.

B Theoretical Analysis

This appendix records the main structural properties used to interpret CARAT. These results are not presented as a complete robustness guarantee against arbitrary adaptive attackers. Instead, they clarify why the method decomposes into a normalized evaluation geometry, a task-aware utility objective, and a constrained robust aggregation step.

Proposition B.1 (Feasibility and support of the capped simplex)**.** *Let*

$$\mathcal{A}(\hat{\rho}) = \left\{ \alpha \in \mathbb{R}^n : \alpha_i \geq 0,\ \sum_{i=1}^{n} \alpha_i = 1,\ \alpha_i \leq \frac{1}{(1-\hat{\rho})n} \right\},$$

with $\hat{\rho} < 1$. Then $\mathcal{A}(\hat{\rho})$ is non-empty, and every $\alpha \in \mathcal{A}(\hat{\rho})$ satisfies

$$|\operatorname{supp}(\alpha)| \geq (1-\hat{\rho})n.$$

Proof sketch. Let $u = 1/((1-\hat{\rho})n)$. Since $\hat{\rho} < 1$, we have $u \geq 1/n$, so the uniform weight vector is feasible and the set is non-empty. For the support bound, if only k coordinates of α are positive, then

$$1 = \sum_{i=1}^{n} \alpha_i \leq ku = \frac{k}{(1-\hat{\rho})n},$$

which implies $k \geq (1-\hat{\rho})n$. Therefore CARAT can suppress suspicious clients, but cannot collapse the aggregate onto an arbitrarily small subset of participants. □

Proposition B.2 (Softmin bounds and gradient form)**.** *Let $s \in \mathbb{R}^T$ and define*

$$\operatorname*{softmin}_{\beta}(s) = -\frac{1}{\beta} \log \left(\frac{1}{T} \sum_{t=1}^{T} e^{-\beta s_t} \right).$$

Then

$$\min_t s_t \leq \operatorname*{softmin}_{\beta}(s) \leq \min_t s_t + \frac{\log T}{\beta}.$$

Moreover,

$$\lim_{\beta \to \infty} \operatorname*{softmin}_{\beta}(s) = \min_t s_t, \qquad \lim_{\beta \to 0} \operatorname*{softmin}_{\beta}(s) = \frac{1}{T} \sum_{t=1}^{T} s_t.$$

Moreover, if $z(\alpha) = C^\top \alpha$ and

$$\pi_t(\alpha) = \frac{\exp(-\beta z_t(\alpha))}{\sum_{\ell=1}^{T} \exp(-\beta z_\ell(\alpha))},$$

then

$$\nabla_\alpha \operatorname*{softmin}_{\beta}(C^\top \alpha) = C\pi(\alpha).$$

Proof sketch. Let $m = \min_t s_t$. Since

$$e^{-\beta m}/T \leq \frac{1}{T} \sum_{t=1}^{T} e^{-\beta s_t} \leq e^{-\beta m},$$

applying $-(1/\beta)\log(\cdot)$ yields

$$m \le \operatorname*{softmin}_{\beta}(s) \le m + \frac{\log T}{\beta}.$$

The limit $\beta \to \infty$ follows immediately from the vanishing gap bound. For $\beta \to 0$, a first-order expansion of the exponential average gives

$$\frac{1}{T}\sum_{t=1}^{T} e^{-\beta s_t} = 1 - \beta \frac{1}{T}\sum_{t=1}^{T} s_t + o(\beta),$$

so the softmin converges to the arithmetic mean. Differentiating the softmin expression with respect to z_t yields the Gibbs weights $\pi_t(\alpha)$, and the chain rule then gives $C\pi(\alpha)$. Hence the ascent direction is a convex combination of certificate columns, with more weight placed on tasks whose current aggregate utility is smaller. □

Proposition B.3 (Directional invariance of certificate scoring)**.** *Suppose two nonzero clipped updates satisfy $\bar{\Delta}_i = a\bar{\Delta}_j$ for some scalar $a > 0$. Then their normalized versions in Eq.* (3) *coincide, i.e.,*

$$\tilde{\Delta}_i = \tilde{\Delta}_j.$$

Consequently, their hidden-task certificates are identical:

$$c_{i,t} = c_{j,t} \qquad \textit{for all } t \in \{1, \dots, T\}.$$

Proof sketch. Eq. (3) rescales every nonzero clipped update to the common radius q_r. Positive collinearity is therefore removed by normalization:

$$q_r \frac{\bar{\Delta}_i}{\|\bar{\Delta}_i\|_2} = q_r \frac{a\bar{\Delta}_j}{\|a\bar{\Delta}_j\|_2} = q_r \frac{\bar{\Delta}_j}{\|\bar{\Delta}_j\|_2}.$$

Substituting the resulting equality into Eq. (4) gives identical task scores. Thus, after clipping, the certificate stage depends on direction rather than residual norm magnitude. □

Proposition B.4 (Task-wise standardization preserves within-task ordering)**.** *Fix a task t and define a standardized certificate score*

$$\hat{c}_{i,t} = \frac{c_{i,t} - m_t}{s_t + \varepsilon},$$

where $m_t \in \mathbb{R}$ and $s_t > 0$ depend only on task t and not on client i. Then for any clients i, j,

$$c_{i,t} \ge c_{j,t} \quad \Longleftrightarrow \quad \hat{c}_{i,t} \ge \hat{c}_{j,t}.$$

Moreover,

$$\hat{c}_{i,t} - \hat{c}_{j,t} = \frac{c_{i,t} - c_{j,t}}{s_t + \varepsilon}.$$

Proof sketch. For a fixed task t, the map $x \mapsto (x - m_t)/(s_t + \varepsilon)$ is affine with strictly positive slope because $s_t + \varepsilon > 0$. Hence it preserves ordering. The difference identity follows by direct subtraction. Therefore column-wise robust standardization calibrates cross-task scale without changing the within-task ranking induced by the raw certificate scores. □

Proposition B.5 (Temporal prior and projected-iterate invariants)**.** *Define*

$$f(\alpha) = \operatorname*{softmin}_{\beta}(C^\top \alpha) - \lambda_r r^\top \alpha - \lambda_p \|\alpha - \alpha^{\text{prev}}\|_2^2.$$

If $\alpha, \alpha' \in \mathcal{A}(\hat{\rho})$ satisfy $C^\top \alpha = C^\top \alpha'$ and $r^\top \alpha = r^\top \alpha'$, then

$$f(\alpha) - f(\alpha') = -\lambda_p \left(\|\alpha - \alpha^{\text{prev}}\|_2^2 - \|\alpha' - \alpha^{\text{prev}}\|_2^2 \right).$$

In addition, under projected gradient ascent

$$\alpha^{(k+1)} = \Pi_{\mathcal{A}(\hat{\rho})}\left(\alpha^{(k)} + \eta \nabla f(\alpha^{(k)}) \right),$$

every iterate remains in $\mathcal{A}(\hat{\rho})$.

Proof sketch. The first statement follows by direct substitution: if the certificate and rank terms match, the only remaining difference in objective value is the quadratic prior term. CARAT therefore prefers, among utility-equivalent feasible weights, the solution closer to α^{prev}. The second statement is immediate from the projection operator. Hence the optimizer preserves three round invariants throughout: $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$, and $\alpha_i \leq 1/((1-\hat{\rho})n)$. □

Proposition B.6 (Norm bound for the final aggregate)**.** *Suppose each clipped update satisfies* $\|\bar{\Delta}_i\|_2 \leq \tau_r$ *and* $\alpha \in \mathcal{A}(\hat{\rho})$*. Then the final aggregate*

$$\Delta_{\text{agg}} = \sum_{i=1}^{n} \alpha_i \bar{\Delta}_i$$

satisfies

$$\|\Delta_{\text{agg}}\|_2 \leq \tau_r.$$

Proof sketch. Because $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$, the aggregate is a convex combination of clipped updates. By the triangle inequality,

$$\|\Delta_{\text{agg}}\|_2 \leq \sum_{i=1}^{n} \alpha_i \|\bar{\Delta}_i\|_2 \leq \sum_{i=1}^{n} \alpha_i \tau_r = \tau_r.$$

Thus the final server step inherits the clipping radius as a hard norm upper bound. □

Proposition B.7 (Per-round complexity)**.** *Let n be the number of clients, T the number of hidden tasks, m the number of sampled coordinates for rank scoring, S the number of rank subsamples, and K the number of projected-gradient steps. Ignoring constant factors from batch size and model width, the extra defense-side work of CARAT scales as*

$$\textit{probe-model forward passes} + O(Snm + KnT).$$

Proof sketch. CARAT performs one baseline probe evaluation plus n candidate probe evaluations over the T hidden tasks, then computes S rank summaries over m coordinates, and finally executes K projected optimization steps over n weights and T task scores. In the current implementation, the probe-model forward passes dominate wall-clock time, while the rank and optimization stages are lower-order terms controlled by m, S, and K. □

Remark on assumptions. The object $\mathcal{D}^{\text{ref}}$ in the main text denotes the labeled server reference set, while the probe pool $\mathcal{P}$ used during each round is a subsampled working set derived from it. In practice, the server first constructs $\mathcal{P}$ from the full server-visible reference data and then samples hidden tasks from $\mathcal{P}$ at each round. This distinction should remain explicit because it is part of the method's server-side assumption and materially strengthens the setting relative to a fully data-free defense.

C Additional Experimental Details

This appendix records the benchmark design and reporting scope that support the main-text experimental protocol.

C.1 Experimental Objectives

The empirical study is designed to test properties specific to CARAT rather than to benchmark it as a generic outlier filter. In particular, we want to know whether a hidden-task certificate built from labeled server data improves robustness when benign clients are heterogeneous, whether that gain persists on class-rich benchmarks where class-selective failure matters, and how much robustness is purchased per unit of trusted-data budget and runtime overhead.

C.2 Benchmark Suites

Primary class-rich benchmark. The main benchmark is CIFAR-100 with ResNet18. This setting best matches CARAT's motivation because the label space is rich enough for different hidden tasks to emphasize different semantic slices, and because average clean accuracy alone can hide substantial per-class degradation.

Scalability benchmark. TinyImageNet with ResNet18 is used to test whether the same aggregation rule remains effective as the number of classes and the visual vocabulary increase. This benchmark is particularly useful for checking whether the hidden-task construction and reference-budget design remain meaningful beyond CIFAR-100.

Application-oriented benchmark. We include CHMNIST with ResNet18 as an application-style benchmark in the appendix. Its role is not to maximize scale, but to show that class-selective failures have a practical interpretation in a domain where server-side auditing with a small labeled reference set is plausible.

Sanity-check benchmark. CIFAR-10 is retained only as a sanity-check benchmark. It remains useful for verifying that CARAT does not fail on standard low-cardinality image classification, but it is not treated as a primary source of evidence because its ten classes are too few to fully exercise the method's class-aware design.

C.3 Threat Models and Attack Selection

Untargeted poisoning. The main untargeted suite contains ALIE, MinMax, and MinSum. Together they represent adaptive model-poisoning attacks that aim to remain stealthy under server-side defenses. FangAttack is evaluated only in the appendix as a stress test, since it is especially informative for geometry-based defenses and should not dominate the design of CARAT's main evaluation.

Targeted and backdoor attacks. The main targeted suite contains ModelReplacement [8], DBA [9], and Neurotoxin [11]. These attacks are better aligned with CARAT's intended advantage because they can preserve global plausibility while damaging specific semantic regions or maintaining durable triggers. BadNets can be reported as an auxiliary sanity-check attack in the appendix, but it is not the main stress case.

C.4 Baseline Families

Untargeted defenses. For untargeted poisoning, the primary baselines are Mean, TrimmedMean, MultiKrum, FLTrust, and CARAT. This set spans plain averaging, coordinate-wise robust statistics, geometry-based filtering, and trusted-data anchoring. Methods such as FLDetector or TriGuardFL can be added in the appendix when the goal is to compare against historical-trajectory or class-wise screening ideas, but they are not required in the core main-text comparison.

Backdoor defenses. For targeted and backdoor attacks, the primary comparison set is Mean, FLTrust, FLAME, DeepSight, and CARAT. These baselines better reflect the representation-level and clustering-based defenses commonly used in backdoor settings than purely Byzantine-oriented rules do.

C.5 Metrics, Fairness, and Reporting

No-attack clean utility is reported separately from attacked runs. The clean metrics are test accuracy and test loss. On class-rich datasets we additionally report macro-average accuracy and worst-class accuracy, because the motivation for CARAT is precisely that averaging can hide semantically concentrated failures. For backdoor experiments, ASR is always reported together with clean accuracy. Runtime per round and total defense overhead are reported for all defenses that add nontrivial server-side computation.

Trusted-data fairness is controlled explicitly. CARAT and FLTrust are compared under the same labeled server-reference budget, rather than allowing one method to benefit from more public data.

In addition to fixed-budget comparisons, we sweep the reference budget in the appendix to show whether CARAT's advantage comes from the hidden-task validation principle rather than from budget alone.

C.6 Shared Training Protocol

Unless stated otherwise, the shared protocol uses 20 clients with full participation, 300 communication rounds, 4 local epochs, and learning rate 0.05. The primary heterogeneity settings are i.i.d. data and Dirichlet non-i.i.d. partitions with $\alpha \in \{0.5, 0.1\}$. The main attacker ratio is 20%, while 40% is reserved for appendix stress tests. The primary experiments use 3–5 random seeds per configuration. Scaling studies in the appendix may additionally vary the number of clients or the attacker ratio, but these are treated as stress tests rather than as the core evaluation axis.

C.7 CARAT Hyperparameters and Ablation Plan

The current implementation defaults are 8 hidden tasks, 4 probe samples per class, probe batch size 128, probe-radius quantile 0.5, MAD clipping with $k = 2.5$, default attacker prior $\hat{\rho} = 0.2$, certificate temperature $\beta = 12.0$, rank weight 0.20, prior weight 0.05, 4096 sampled coordinates, 5 rank subsamples, 60 projected optimization steps, and optimizer learning rate 0.5.

Two design principles matter when transferring these defaults across datasets. First, the reference budget should scale with the number of classes rather than be fixed as a single dataset-independent total pool size. Second, CARAT should be ablated along the axes that correspond directly to its mechanism: hidden-task count, reference budget per class, rank-penalty weight, temporal-prior weight, and clipping controls. The appendix therefore includes ablations that isolate whether the observed robustness comes from class-balanced hidden tasks, from the structural rank penalty, or simply from access to more trusted data.